

Memo: A framework for modeling functional interdependencies within organizations

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Abstract

In the formal modeling literature, interdependencies between organizational components manifest through their impact on a unified measure of performance; the contribution to performance of a single component depends on the state of the components with which it has interdependencies. Interdependencies, when formulated as such, are superficial because changing the state of a component does not affect the state of other components, only the fitness contributions. In this paper, I propose a model of functional dependencies based on the well-known spin-glass model, in which interdependencies are modelled holistically by taking a matrix of interdependencies and yielding probabilities for entire configurations. While these functional interdependencies can be modeled alongside the traditional fitness interdependencies, an analysis of functional interdependencies alone provides fresh insights into the imperfect process of organizational configuration; common assumptions about forward causation imply that returning to previous locations is extremely costly and therefore search becomes a highly path dependent walk on the energy surface.

Introduction

Understanding the context for action is essential in any behavioral science and over the last 25 years the study performance landscapes has done much to fill in gaps around our collective understanding of the context for action in organizational theory. Levinthal [1997] introduced the notion of a performance, or fitness, landscape to the management science community and by 2019 a review by Baumann et al. [2019] identified 70 simulation based papers in top journals

using some version of the NK model Levinthal originally imported from evolutionary biology Kauffman and Weinberger [1989].

The NK model has proven so useful because it parameterizes the degree of interdependencies in the choices that managers make. Organizations often face complex problems in which one choice affects the value of other choices Simon [1962]. Before the introduction of the NK model, the consequences of interdependencies for how individuals and organizations approach decision-making was, almost by necessity, unstudied. The NK model, however, elegantly linked the interdependent choices to performance values by creating a fitness landscape which showed that interdependencies tended to create “rugged” landscapes. Ruggedness refers to the count of local fitness optima, that exist when the change of any single characteristic results in a lower fitness value. Ruggedness renders boundedly rational actors ineffective at searching complex landscapes and given this the literature focuses on how to improve the effectiveness of search (see Baumann et al. for an excellent summary of the literature).

As important as this literature has been, the implementation of interdependencies is superficial in the sense that changing one component only alters the *fitness contribution* of other components, not the state of components themselves. The distinction matters greatly for how we think about organizational learning and problemistic search.

Consider the case of the Boeing 737 Max plane, grounded for over 18 months after two fatal crashes and the several near crashes. The saga of the 737 Max reveal something fundamental about the nature of organizational decision-making; changes to state of one component, whether in a product or the organization itself, often alters the state of other components. In the case of the 737 Max, Boeing was facing superior competition in the narrow-bodied aircraft segment from Airbus, whose A320neo model was significantly more fuel efficient. To avoid losing the whole market segment, Boeing needed to incorporate more fuel-efficient engines, which happened to be larger. This wasn’t a problem for the A320 models, but the 737 models were shorter and couldn’t accommodate the engines in the usual way.

Faced with the prospect of a decade long design and certification process for a whole new plane that broke interoperability standards with the 737 series, Boeing chose to create version that worked around the problem of the engine size. Instead of being placed directly under the wing like prior models, the larger engines were placed far enough forward on the wing to be raised closer to flush with the wing. It was known that this would require rebalancing the weight, but it was not understood immediately that doing so would create shock waves over the wings at

high speed and provide extra lift around the engines¹.

This led to some behaviors that even seasoned pilots would not expect. This is against FAA rules so Boeing introduced a new computer control system, a new external sensor and optional override switches. Pilots were not to be trained on the operation of these systems in simulators in order to avoid incurring penalties for creating additional costs for the airlines purchasing the aircraft. The end result was 346 deaths, severe reputational costs for Boeing and the FAA, fines, lost revenue, and more; an unmitigated disaster.

While many case studies will be written on this sequence of events and its causes, it highlights the fact that altering an aspect of a complex system often alters other aspects, *not just the performance contribution of the other components*. Decision makers might be able to choose the state of a focal component exactly as they wish, but they do not ultimately choose the position of the whole entity in the configuration space. Functional interdependencies or correlations between state variables require that the changing of one state alters the others.

This is not how search is currently studied in the performance landscape literature. Instead, the position in the state space is fully controlled by the decision maker and it is the fitness of the position that is unknown. To understand why the NK models interdependencies this way, I revisit the origin of the model.

Interdependencies in the NK Landscape

In late 2020, Google's AlphaFold announced it had solved the "protein folding problem" of accurately predicting the atomic structure into which proteins will fold based solely on the sequences of amino acids in the protein. This was one of the "grand challenges" of biology because of the sheer number of atomic and thermodynamic interdependencies between the amino acids constituting the protein. In the context of this breakthrough, it is worth revisiting Kauffman's NK model of fitness landscapes because protein folding was at the core of the problem the model sought to address Kauffman and Weinberger [1989], Kauffman et al. [1988].

Kauffman's model was originally concerned with the dynamics of the evolution of antibodies in immune responses. A well-adapted antibody—a folded protein—can bind itself to a pathogen, or rather a surface on the pathogen labeled the *antigen*, and destroy or otherwise disable the pathogen. Thus, in order to understand the dynamics of immune response, one needs to

¹<https://www.seattletimes.com/seattle-news/times-watchdog/the-inside-story-of-mcas-how-boeings-737-max-system-gained-power-and-lost-safeguards/>

understand how mutations within the antibody system can search over the space of possible proteins to find ones that have the high *affinity* necessary for bonding with the relevant antigen. The ability of a specific antibody protein to bind with the antigen is directly related to the body's ability to have a robust immune response. This means the affinity, or binding strength, is a form of fitness within the scope of the problem and the fundamental question is "what are the dynamics of the immune system response as it searches for successfully binding molecules?" Given that the immune system cannot solve the protein folding problem, it appears to make use of random mutations in antibody DNA to attempt to find good matches. The dynamics of search then are determined by the structure of the "affinity landscape" of protein bonding.

When Kauffman notes that this affinity landscape is unknown, "although knowable" Kauffman and Weinberger [1989][211], this is a reference to the protein folding problem. In the absence of a known landscape, Kauffman suggests that we can characterize the features of the unknown landscape by parameterizing the degree of interdependencies between amino acids. When there are no interdependencies, one can alter amino acids one at a time to find a protein with maximum affinity; altering any individual amino acid (or allele in genetics-oriented interpretations of the model) does not impact the *contributions* of the other amino acids. As interdependencies increase, however, changing any one amino acid necessarily alters the contributions of others by changing the atomic structure of the resultant folded protein.

Because of the complexity of the protein folding problem, there was no principled reason for choosing one pattern of interdependencies over another and Kauffman developed the NK model as a random null model of the effects of interdependencies of a given degree. The consequences of this choice and some of its insufficiencies when it comes to applying the model to the management science has been addressed elsewhere.

This random-null specification was a very defensible choice on two different levels, given the challenge of the protein folding problem. The first level is the protein folding problem itself: at the time, it was not possible to predict the structure of the protein that results from changing any given subset of amino acids. The second level is knowing the fitness of the resulting whole. If one knows the structure of the resulting protein, then the fitness or affinity value of the protein can be assessed relative to a given antigen. Kauffman and his collaborators had no means of knowing the protein structure and therefore the fitness of the protein. Given this, the NK model blackboxes the folding problem and links expected fitness scores directly to the degree of interdependencies between amino acids in various position on the strand. This approach made

it possible to model the effects of interdependencies on both the resulting fitness landscape and immunological search over the landscape. Crucially, the model does not consider the features of the whole protein produced through changes, only the resulting fitnesses.

A strictly evolutionary learning process does not need more than fitnesses either. Without an inability to create abstracted representations of the nature of the protein folding problem, immunological search can only make use of the fitness outcomes to further direct search. This is what all agents on an NK landscape do. Humans and our various epistemological systems *do* try to represent the first problem, however. Just as tremendous resources have been directed at solving the protein folding problem, professional sports teams, for example, pour millions of dollars into creating hyper-detailed analytics to try to understand what goes on in the course of the game, what players, types of players, combinations of players, training and rest regimes, and a host of other considerations contribute to the ultimate success of the team. Coaches and team management do not simply make random perturbations to personnel and look at the final score or record.

Representing the internal state of an organization, the equivalent of the folded protein, is essential to organizational learning and decision making. While the digitization of much of organizational life has allowed for a now thriving industry of inward-facing business and people analytics, the effort to represent the internal state has always been a part of effective management, whether through management education, popular lay theories or practitioner's individual sensemaking.

The fact that humans try to represent the internal state of an organization in the process of incorporating feedback about that state from the external environment affects the learning process, particularly when there are interdependencies across the various controllable attributes of the organization. In this paper, I re-envision search on fitness landscapes in light of this claim and argue the differences affect the dynamics and outcomes.

A model of functional dependencies

To model functional interdependencies, I use the insights of the Ising model of ferromagnetism as crucial building blocks. It has a variety of desirable features that I will discuss as they are introduced. Crucially, this model can be laid over any fitness landscape for a configuration model, like the NK model or Rahmandad's PN model Rahmandad [2019]

The Ising, or spin glass, model represents atoms with dipole magnetic moments as embedded in graph of interdependencies or interactions. Atoms have a positive and negative pole and have an orientation or spin. When the spins of the atoms point in different directions, the graph of atoms is a disordered “glass”. However, when the spins point in the same direction, the atoms collectively exhibit a magnetic field, i.e. become a magnet. In this sense, the Ising model is a configurational model; the primary question it addresses is the conditions under which the atoms can become configured in two particular states (all up or all down) from among the combinatorial huge number of possible configurations.

Such aligned states are not important to the present exercise, but the model offers the mathematical machinery necessary for modeling functional interdependencies and the transition probabilities between the configurations. To show this, the first essential insight is that we can treat the components of configurational models like the NK model as nodes in a network with two different spin states. Typically those are 0, 1, but there is no loss of generality to relabel them as -1 and 1 to correspond to the spins within the Ising model.

With this relabeling, the NK model is a Ising model with no ties and therefore no interactions between the states of the individual components. To introduce such interactions, we add ties between the nodes. We can express all the functional dependencies between components using a symmetrical matrix J in which there is a row and column for each component and the cell value is the first order strength of *attraction* between components. For example, below is an interdependencies matrix J with $-1 < J_{i,j} < 1$.

J	1	2	3	4	5
1	1.0	.40	.70	.35	0.0
2	.40	1.0	-.5	.94	.12
3	.70	-.5	1.0	.90	-.7
4	.35	.94	.90	1.0	0.0
5	0.0	.12	-.7	0.0	1.0

Each component is maximally attracted to its own state, hence the 1's down the diagonal. Zero means there is no state attractions or dependency between the components. To be clear, $J_{i,j}$ is not the probability that component i and component j are in the same state because ultimately that will depend upon the current state of the whole system and other aspects of the model. The values instead describe the independent attraction between the two states of the bits, which is a crucial, but not exclusive, input into understanding the underlying transition probabilities

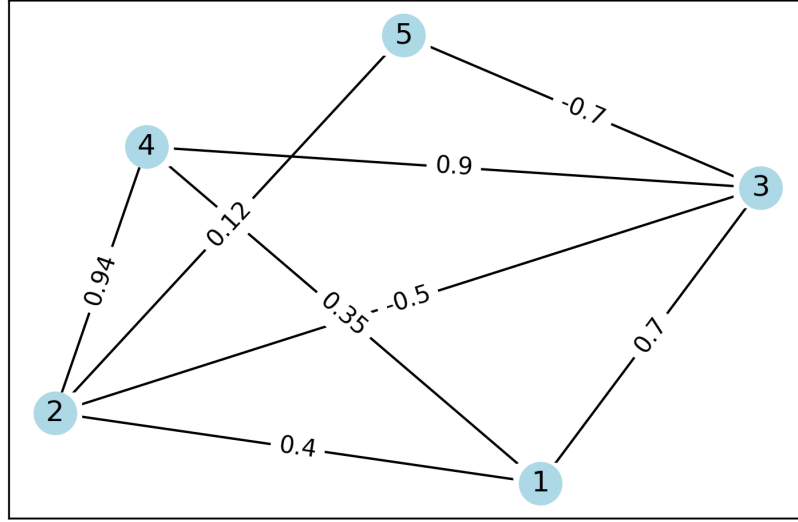


Figure 1: The graphic depiction of a configurational model with interaction matrix J . Nodes are the individual components and the ties represent the strength of interaction between the components.

between configurations.

To better understand this, consider Figure 1. If you look closely at components 2, 3 and 4, you'll see that nodes 2 and 3 have a negative attraction, but 2 & 4 and 3&4 have a strong positive attraction. This presents a tension for the whole system because 2 and 3 want to be in opposite orientations, yet 4 wants to be in the same oriented of both 2 and 3. Let us say that the current state of the system/configuration of components is $[1, 1, -1, -1, 1]$. There is tension because 2 and 4 want to take on the same value but currently have different ones. Now if an actor changes component 2 to value -1, that tension between 2 and 4 is resolved but it creates a new tension between 2 and 3.

For us, the question becomes the following: If we allow the interdependencies to dictate a final configuration based on the change, what configuration should we expect the system to take on? The Ising model offers a way to answer this question probabilistically by assigning probabilities to configuration based inversely related to tensions.

The Ising model does this through the use of the Hamiltonian of the system, which is a widely used equation for calculating the total energy of a physical system. We now define it.

Each component $k \in \Lambda$ has a spin $\sigma_k \in \{+1, -1\}$. In the NK model, we usually think of the component as being in state $\{0, 1\}$, but the change is trivial as far as the usual interpretation goes.

(It's just two labels in the NK model and could be A and B.) It does matter for the calculation obviously and we will need to think about different possible values and their relationship to the interpretation.

A spin configuration—a location in the hypercube of the configurational model—is denoted as $\sigma = (\sigma_k)_{k \in \Lambda}$

Given any two components $i, j \in \Lambda$, the matrix $J_{i,j}$ defines the interactions between individual components. The canonical model also has a matrix H_i that defines an external magnetic field the components interact with and a magnetic moment μ of that field. This external field offers a means to model a variety of external influences on the final configuration, but that is outside the scope of the current short paper. The Hamiltonian $H(\sigma)$ of the configuration σ is thus,

$$H(\sigma) = - \sum_{\langle i,j \rangle} J_{i,j} \sigma_i \sigma_j - \mu \sum_j h_j \sigma_j \quad (1)$$

Physical systems “want” to minimize energy, so the lower the total energy of a configuration, the higher probability the configuration. But external energy in the form of heat can moderate this because heat overwhelms the interactions. These insights are combined in the Boltzmann distribution. The Boltzmann probability of configuration σ is

$$P_\beta(\sigma) = \frac{e^{-\beta H(\sigma)}}{Z_\beta} \quad (2)$$

where $T \in (0, \infty)$ is the temperature, k_b is the Boltzmann constant, $\beta = \frac{1}{k_b T}$ and

$$Z_\beta = \sum_{\sigma} e^{-\beta H(\sigma)} \quad (3)$$

This framework allows us to start with a matrix of functional interdependencies and understand the ways in which the tensions present in the particular set of interdependencies are likely lead to particular configurations as individual changes are made.

Because we are not modeling an atomic system, we can drop the Boltzmann constant in the calculation of probabilities. Doing so doesn't alter the range of possible temperatures, but makes them more directly interpretable. $T \in (0, \infty)$ but $T \rightarrow 0$ gives us the frozen system in which only the lowest energy states have any probability mass. But, for the small systems we are looking at, $T \approx 1000$ makes all states equiprobable. Table 1 gives the Boltzmann probabilities for several different temperatures for the example interdependencies matrix J from above.

Configuration	T=.1	T= .5	T=10	T=1000
(00000)	0.0	0.0018	0.0295	.03125
(00001)	0.5	0.0018	0.0295	.03125
(00010)	0.0	0.0018	0.0295	.03125
(00011)	0.0	0.0018	0.0295	.03125
(00100)	0.0	0.0018	0.0295	.03125
(00101)	0.0	0.0005	0.0275	.03125
(00110)	0.0	0.0111	0.0322	.03125
(00111)	0.0	0.0027	0.0301	.03125
(01000)	0.0	0.0018	0.0295	.03125
(01001)	0.0	0.0023	0.0298	.03125
(01010)	0.0	0.012	0.0324	.03125
(01011)	0.0	0.0152	0.0328	.03125
(01100)	0.0	0.0007	0.028	.03125
(01101)	0.0	0.0002	0.0264	.03125
(01110)	0.0	0.0267	0.0337	.03125
(01111)	0.0	0.0084	0.0318	.03125
(10000)	0.0	0.0018	0.0295	.03125
(10001)	0.0	0.0018	0.0295	.03125
(10010)	0.0	0.0037	0.0305	.03125
(10011)	0.0	0.0037	0.0305	.03125
(10100)	0.0	0.0074	0.0316	.03125
(10101)	0.0	0.0018	0.0295	.03125
(10110)	0.0	0.0002	0.0358	.03125
(10111)	0.0	0.0223	0.0334	.03125
(11000)	0.0	0.0041	0.0307	.03125
(11001)	0.0	0.0052	0.031	.03125
(11010)	0.0	0.0537	0.0349	.03125
(11011)	0.0	0.0683	0.0353	.03125
(11100)	0.0	0.0061	0.0313	.03125
(11101)	0.0	0.0019	0.0295	.03125
(11110)	0.5	.485	0.0389	.03125
(11111)	0.0	.152	0.0367	.03125

Table 1: The probabilities of all configurations (row) for the given temperature T (columns). Near $T = 0$, only the lowest energy configuration have any probability of materializes. As T grows, all configurations tend toward being equally probable. Note that two possible states of each node is $\{0,1\}$. If the states were $\{-1,1\}$, each location and its negation would be equiprobable. More generally, the possible states of each component are meaningful within the model.

Conditional probabilities and behavior

As an actor considers configurations within a system with functional interdependencies, it does not necessarily know what the end configuration will be after changes to any number of changes to individual components. Search becomes a problem of learning with states are likely and high

performing. This search can be modeled by calculating the probability of states conditional on particular components being placed in particular states. After the actor specifies the particular changes, a conditional Boltzmann probability and a temperature is used to determine the final configuration.

Initial analyses of this search process reveals the vital importance of temperature in the presence of functional interdependencies. When the temperature is low, the problem landscape is effectively reduced to a small subset of possible points and actors can easily search these and maximize performance. However, as temperature increase and all other states become possible, actors often cycle through regions of the configuration space endlessly because they lack control over the final configuration. As temperatures increase significantly, all states are equiprobable and interdependencies fade in importance (but transitions are stochastic).

Particularly for mid-range temperatures, *the model highlights the imperfect nature of search for good configurations when there are functional instead of just fitness dependencies*. The formal literature has lacked this nuance and the Ising framework offers other intriguing levels to enrich the formal study of the messy work of finding workable and high performing configurations.

Relationship to Configurational Research

The source inspiration of this model is the performance landscape literature and, in particular, recent research (Atwell and Apaza, working paper) that demonstrates that the “ruggedness” created by a high degree of interdependencies is likely not much of an impediment to effective search, as has widely been assumed to be the case in the literature. But human actors do seem to struggle with search and the question of why remains. Functional dependencies as modeled here appear to present a challenge to search. It was only through the process of researching how others conceive of interdependencies that the author realized there might be an important connection between the patterns this model creates and the notion of “limited diversity” discussed in the configurational approach; functional interdependencies and a tendency of actors to take small “steps” in combinatorial/configurational space could explain limited diversity because many configurations adjacent to the focal one could low probability states and an actor’s attempt to change the those configurations will be thwarted by the functional dependencies between characteristics.

To use Ragin’s example from the foreward of the RSO volume *Configurational Theory and*

Methods in Organizational Research(pp. XVIII), countries can be described as having (or not) “corporatist relations”, “strong unions” and “strong leftists parties”. These attributes appear functionally interdependent given the limited diversity. The limited diversity observed in this setting includes many cases of countries having all three or none of the attributes (11 total) and only 5 with mixed states. The application of the spin-glass model to the dependency matrix below would predict this limited diversity in the sense that having all three attributes or their negation is the energy minimum of the system and both are the most probable states. Other, mixed states are less likely, but observable from time to time. For example, three of the mixed cases do not have strong left parties and strong unions, suggesting those two attributes have a strong functional dependency which is represented with an attraction strength of .7 in the matrix. The other values in the matrix accommodate the remaining two mixed cases but also strongly predict the present or absent of all three attributes.

<i>J</i>	SLP	SU	CR
SLP	1.0	.70	.15
SU	.70	1.0	.40
CR	.15	.40	1.0

The model predicts these outcomes before one starts to incorporate search behavior, but again, the current working thesis of the author is that the structure of the environment in the form of functional interdependencies is going to strongly guide search processes and outcomes. This connection between the literature on search and configurational research is something the author looks forward to exploring more.

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